LEARNING RESOURCE GUIDE

Generative AI for Beginners | Professional Participant Workbook

Section	Section 1: Introduction to Generative AI
Lecture	Lecture 1.3: Understanding Machine Learning Basics

Learning Objectives

By the end of this lecture and its accompanying guide, you will be able to:

- Explain how machine learning works using the analogy of learning from examples.
- Distinguish between supervised, unsupervised, and reinforcement learning with concrete examples.
- Describe what training data is and why its quality and quantity are critical to ML system performance.
- Identify how each ML type is applied in real-world professional and business contexts.
- Articulate key failure modes for each type of machine learning and how to mitigate them.

Introduction

Machine learning is the engine that powers nearly everything in modern AI — including all generative AI systems. Yet the term is often used as though it explains something, when in fact it is the starting point of a deeper question: how exactly does a computer system learn from data? What does learning mean for a machine? And why does the quality of that learning depend so heavily on the data it receives?

The three types of machine learning — supervised, unsupervised, and reinforcement learning — are not interchangeable. They address different problems, require different data, and are applied in very different contexts. Supervised learning produces the spam filters and diagnostic systems that classify labelled inputs. Unsupervised learning discovers hidden structure in unlabelled data. Reinforcement learning trains agents to make sequences of decisions through reward-based feedback.

Equally important is understanding the role of data. Machine learning systems are fundamentally data-driven — their performance ceiling is set by the data they learn from. A system trained on incomplete, biased, or unrepresentative data will fail in predictable ways. For professionals making decisions about AI adoption, data stewardship is as important a topic as algorithm selection.

Core Concepts Explained

Supervised Learning

Definition: Supervised learning trains a model on labelled data — pairs of inputs and correct outputs — so the model can predict labels for new, unseen inputs. The supervision comes from the labelled training examples.

Business relevance: Supervised learning powers the majority of currently deployed commercial AI applications: fraud detection, medical image analysis, product recommendation, customer churn prediction, and document classification. It requires substantial investment in data labelling.

Practitioner relevance: For professionals overseeing AI projects, supervised learning implies that accurate, representative, and correctly labelled training data is the primary determinant of system quality. Budgeting for data labelling is as important as budgeting for model development.

Real-world example: A bank fraud detection model is trained on millions of historical transactions, each labelled as fraudulent or legitimate. After training, the model can assess a new transaction in milliseconds and assign a probability that it is fraudulent — a task that previously required manual review.

Risks or limitations: If the labelled training data reflects past human decisions that were biased — for example, historically higher fraud flags for certain geographies — the model will perpetuate and potentially amplify those biases.

Unsupervised Learning

Definition: Unsupervised learning identifies patterns and structures in unlabelled data without pre-specified correct answers. The model discovers categories, clusters, or representations on its own.

Business relevance: Unsupervised learning is used for customer segmentation, anomaly detection, dimensionality reduction, and topic modelling. It is valuable when labelling data at scale is impractical or when the structure of the data is unknown in advance.

Practitioner relevance: Professionals who understand unsupervised learning can recognise when it is being applied and what its outputs mean. Cluster labels from unsupervised models are algorithmically defined — they require human interpretation to carry business meaning.

Real-world example: A retail company runs a clustering algorithm on customer purchase histories and discovers five distinct customer segments without pre-defining those categories. Marketing campaigns can then be tailored to each segment's observed behaviours.

Risks or limitations: Clusters found by unsupervised algorithms reflect patterns in the data, but those patterns may be artefacts of data collection methods rather than meaningful real-world categories. Careful validation is required before acting on them.

Reinforcement Learning

Definition: Reinforcement learning trains an agent to make sequential decisions in an environment by rewarding correct behaviours and penalising incorrect ones. The agent learns a policy — a mapping from states to actions — to maximise cumulative reward.

Business relevance: Reinforcement learning is used in robotics, game-playing AI, supply chain optimisation, and increasingly in fine-tuning large language models via Reinforcement Learning from Human Feedback — RLHF. It is the mechanism by which ChatGPT was fine-tuned to be conversational and helpful.

Practitioner relevance: Understanding that reinforcement learning requires an environment with a well-defined reward signal is important for scoping RL applications. Poorly designed reward functions can produce systems that achieve the specified reward through unintended means — a failure mode called reward hacking.

Real-world example: AlphaGo learned to play Go at superhuman levels using reinforcement learning — not by being given rules, but by playing millions of games and receiving reward signals when it won. The same principle, applied to text generation with human feedback as the reward, underlies RLHF in modern language models.

Risks or limitations: Reward hacking is a documented failure mode: the agent finds ways to maximise the reward signal that violate the intent. An RL agent optimising for user engagement might learn to promote outrage-inducing content if that drives more clicks.

Training Data Quality and Quantity

Definition: Training data is the dataset on which a machine learning model is trained. Quantity affects what patterns the model can learn; quality determines whether those patterns are accurate and representative of the target domain.

Business relevance: Data is the primary input to machine learning. Organisations that have invested in high-quality, well-labelled, domain-specific data have a structural advantage in developing effective ML systems. Data is often the key differentiator between competitive AI products.

Practitioner relevance: For any professional overseeing an AI initiative, understanding data requirements — volume, labelling, representativeness, recency, bias risk — is as important as understanding the algorithm. Data governance, documentation, and quality control are non-negotiable.

Real-world example: A hospital with twenty years of annotated medical imaging records, with physician-confirmed diagnoses linked to each image, is in a fundamentally stronger position to train a medical AI system than an organisation relying on publicly available, less carefully annotated datasets.

Risks or limitations: Common data quality issues include class imbalance (rare events underrepresented), label noise (incorrect labels), distribution shift (training data from a different context than deployment), and historical bias embedded in labels.

Practical Applications

Supervised Learning in Financial Services

Credit scoring, fraud detection, anti-money-laundering screening, and loan underwriting decisions all use supervised learning. The shared requirement is historical data with reliable outcome labels. The shared risk is that historical outcomes may encode past discrimination, which the model then perpetuates.

Unsupervised Learning in Marketing Analytics

Customer segmentation, product affinity clustering, and topic modelling of customer feedback are common unsupervised applications. These analyses surface patterns that analysts then interpret and act upon. The key professional skill is critically evaluating whether algorithmically discovered segments are stable and actionable.

Reinforcement Learning in Operational Optimisation

Warehouse robotics, traffic light optimisation, HVAC energy management, and data centre cooling are real deployments of reinforcement learning. The design of the reward function is the most critical engineering decision — a poorly specified reward produces behaviour that achieves the letter but not the spirit of the objective.

Best Practices

- Define the problem type before selecting an approach.Determine whether you need to predict a label (supervised), discover structure (unsupervised), or train an agent (reinforcement). The choice determines data requirements and evaluation methods.
- Invest in data quality as seriously as algorithm selection.For ML systems, data quality is typically the binding constraint on performance. A sophisticated algorithm applied to poor data performs worse than a simple algorithm applied to excellent data.

- Evaluate training data for representativeness and bias.Before deploying any ML system, audit the training data for class imbalance, label noise, historical bias, and distribution mismatch with the target environment.
- Plan for distribution shift over time.ML models trained on historical data degrade as the world changes. Establish monitoring and retraining protocols before deployment, not after performance declines are noticed.
- Separate training, validation, and test datasets rigorously.Using the same data for training and evaluation produces overoptimistic performance estimates. Rigorous separation is a foundational requirement of trustworthy ML development.

Common Mistakes and Misconceptions

Misconception: Assuming more data always improves model performance.

Correct approach: More data of poor quality can harm performance. A model trained on more biased data will be more biased. Quality, representativeness, and correct labelling matter as much as volume.

Misconception: Treating the AI system as a black box once deployed and not monitoring it.

Correct approach: ML systems can degrade over time as real-world inputs drift away from the training distribution. Ongoing monitoring, evaluation, and periodic retraining are operational requirements.

Misconception: Believing that a model performing well on training data is ready for deployment.

Correct approach: Performance on training data is not a reliable indicator of real-world performance. Models must be evaluated on held-out test data that was not used during training.

Misconception: Assuming that an AI system is objective because it is mathematical rather than human-made.

Correct approach: ML systems encode human decisions in data labelling, feature selection, reward function design, and evaluation metric choice. Objectivity must be deliberately designed and audited — it is not automatic.

Decision Framework: Selecting the Right ML Approach — Quick Reference

Use this framework when evaluating what type of machine learning approach is appropriate for a given problem.

Question	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Do you have labelled data?	Required	Not required	Not required (reward signal instead)
Do you know the output?	Yes — predefined labels	No — discovered by model	No — agent learns a policy
Primary task	Predict or classify	Discover patterns or structure	Make sequential decisions
Typical applications	Fraud detection, diagnosis, classification	Segmentation, anomaly detection	Robotics, game AI, RLHF fine-tuning
Main data risk	Label bias and class imbalance	Artefactual clusters	Reward hacking
Evaluation method	Accuracy, precision, recall, F1	Cluster quality, human validation	Cumulative reward, human evaluation

How to use this: *When evaluating an AI product, ask the vendor which row of this table describes their approach — and then probe whether the data requirements and risks have been addressed.*

Knowledge Check

Test your understanding before moving on. Answers are shown below each question.

1. A healthcare company wants to build an AI system that flags patient records most likely to result in readmission within 30 days. Which ML type is most appropriate?

A. Unsupervised learning — no labelled data required.

B. Supervised learning — historical patient records labelled with readmission outcomes.

C. Reinforcement learning — a reward function based on hospital costs.

D. Rule-based AI — explicit clinical criteria from physicians.

Answer: B. Supervised learning — historical patient records labelled with readmission outcomes.

Why: *Predicting a specific outcome from historical records with known outcomes is a supervised learning task — it requires labelled training data where the outcome is known.*

2. A retail analyst uses an ML algorithm to group customers by purchase behaviour, with no predefined categories. Three months later, she cannot explain what the group labels mean to her manager. What is the most likely explanation?

A. The algorithm failed and produced random groupings.

B. Unsupervised learning clusters are algorithmically defined and require human interpretation; the analyst did not translate algorithm output into business meaning.

C. Supervised learning was used incorrectly.

D. The training data contained too many records.

Answer: B. Unsupervised learning clusters are algorithmically defined and require human interpretation; the analyst did not translate algorithm output into business meaning.

Why: *Unsupervised learning discovers patterns but does not label them meaningfully. The analyst's role is to interpret the clusters and translate them into actionable business categories.*

3. An RL system trained to maximise click-through rates on an online platform consistently recommends provocative and misleading content. What failure mode does this illustrate?

A. Overfitting to training data.

B. Label noise in the training set.

C. Reward hacking — the system maximised the specified reward through unintended means.

D. Distribution shift between training and deployment environments.

Answer: C. Reward hacking — the system maximised the specified reward through unintended means.

Why: *Reward hacking occurs when an RL agent achieves high reward through behaviours that violate the intent behind the reward function. Click-through maximisation without quality constraints produces this outcome.*

4. A model performs at 97% accuracy on its training dataset but only 62% accuracy when deployed in production. What is the most likely explanation?

A. The model was trained using reinforcement learning.

B. The training dataset was too small.

C. The model was not evaluated on held-out test data and the real-world distribution differs from the training distribution.

D. The model uses unsupervised learning, which does not support accuracy measurement.

Answer: C. The model was not evaluated on held-out test data and the real-world distribution differs from the training distribution.

Why: *High training accuracy with poor deployment performance indicates overfitting and distribution shift — the model memorised training data patterns that do not generalise to real-world inputs.*

Reflection Questions

1. Can you identify a decision in your organisation or industry that is currently made by humans but could potentially be framed as a supervised learning task? What data would be required?
2. Think of a process in your field that involves discovering hidden patterns in large datasets. Would unsupervised learning be applicable? What would you expect it to discover?
3. Reward hacking in reinforcement learning has analogies in human organisations. Can you think of examples where optimising a metric produced unintended consequences?
4. If you were responsible for procuring an ML-based product, what three questions about training data would you ask the vendor?
5. How should organisations balance the commercial pressure to deploy AI quickly with the quality and governance requirements that responsible ML development demands?

Key Takeaways

- Machine learning systems learn patterns from data rather than executing explicit rules — performance is bounded by data quality and representativeness.
- Supervised learning requires labelled data; unsupervised learning discovers hidden structure in unlabelled data; reinforcement learning trains agents through reward-based feedback.
- RLHF is the mechanism used to fine-tune conversational AI systems like ChatGPT — making them helpful, harmless, and honest through human preference signals.
- Data quality — including representativeness, labelling accuracy, and freedom from historical bias — is typically the binding constraint on ML system performance.
- ML models must be evaluated on held-out data not used during training; training accuracy is not a reliable proxy for deployment performance.
- Distribution shift — when the deployment environment differs from the training environment — is a leading cause of ML system degradation over time.
- Reward hacking is a specific failure mode of RL systems where the agent achieves specified reward through means that violate its intended purpose.

Recommended Next Actions

6. For each of the three ML types, write one concrete example from your professional domain where that approach would be applicable.
7. Review a recent AI product or project announcement and identify what type of ML it uses and what that implies about its data requirements and potential failure modes.
8. Add to your glossary: Supervised Learning, Unsupervised Learning, Reinforcement Learning, Training Data, Distribution Shift, Reward Hacking, RLHF.

9. Write three questions you would ask about training data quality before deploying any ML system in your organisation.